



5366 - Maximizing the value of extragalactic JWST surveys using realistic, highly-resolved simulations

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Some of the earliest results from JWST challenge our existing picture of galaxy evolution. These include a claimed overabundance of UV-luminous galaxies at $z > 10$, as well as early-forming massive galaxies and black holes (at $z > 5$), though controversies abound. Robust interpretation of the new data requires a better understanding of the many assumptions that underpin our inference methods and proper modelling of observational biases. Our proposed program leverages hundreds of highly-resolved simulations already run with FIRE physics, as well as state-of-the-art radiative transfer-based forward-modelling techniques developed by our team, to generate a comprehensive suite of synthetic observations, modelled using different assumptions, such as different stellar initial mass functions, stellar population synthesis models, dust modelling, and emission from active galactic nuclei. By training on forward-modelled data in association with the 'ground truth' known from the simulations, we will test and refine fitting tools commonly applied to observations of high-redshift galaxies. We will focus on extracting reliable redshifts, stellar masses and star formation rates

from photometry and spectra. The crucial insights on parameter inference, model degeneracies, and uncertainty estimation resulting from this project will enable us to maximize constraints on models of galaxy formation and the value of JWST. The systematic variation of forward-modeling parameters over the largest set of highly-resolved galaxies available will also provide a publicly-available legacy dataset of synthetic SEDs with immense community value.

OBSERVING DESCRIPTION